

Having fun with RISC-V: Online Appendix, Plots

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I. Execution time against GHC (appendix Table I)

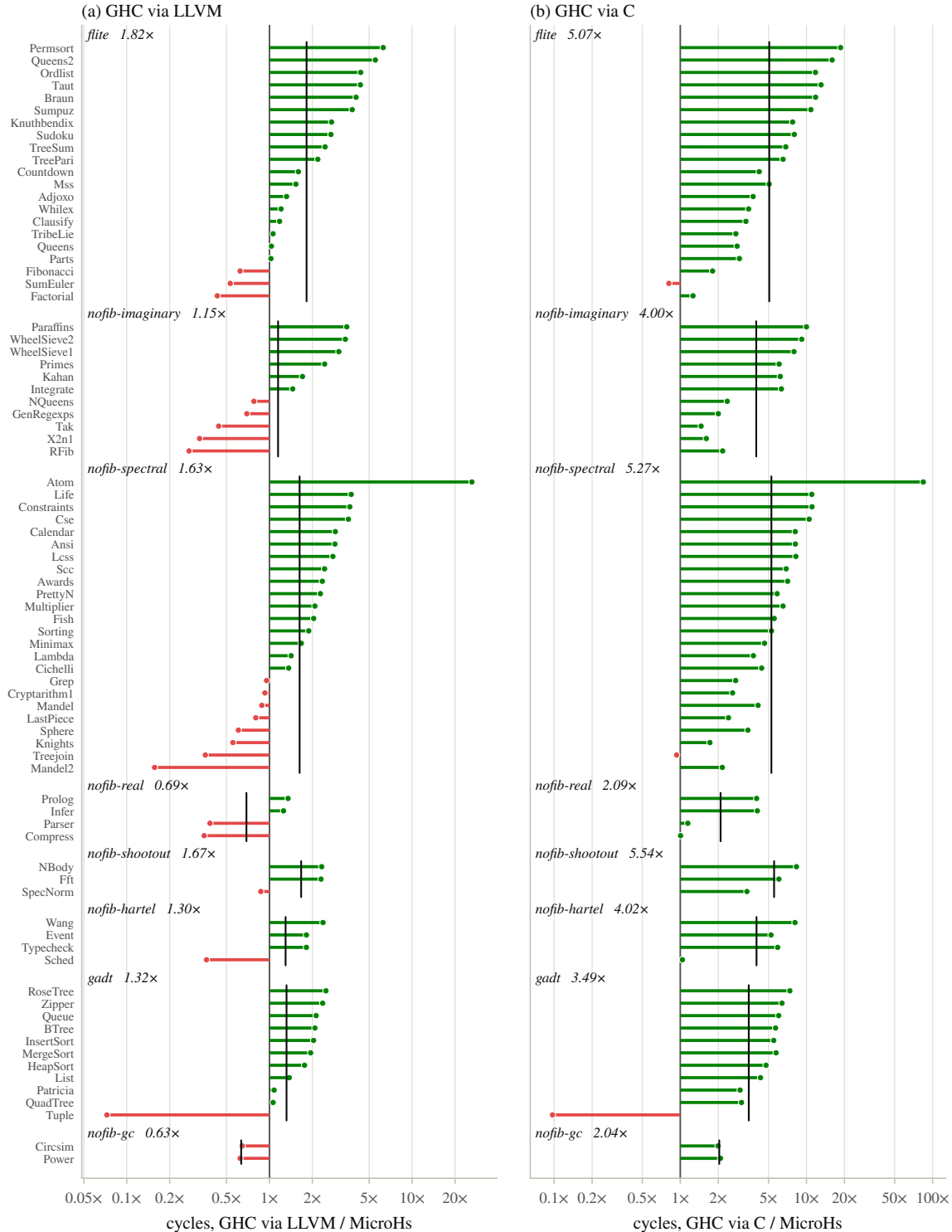


Fig. 1. Cycles, GHC over MicroHs, per program and suite, with suite geometric means.

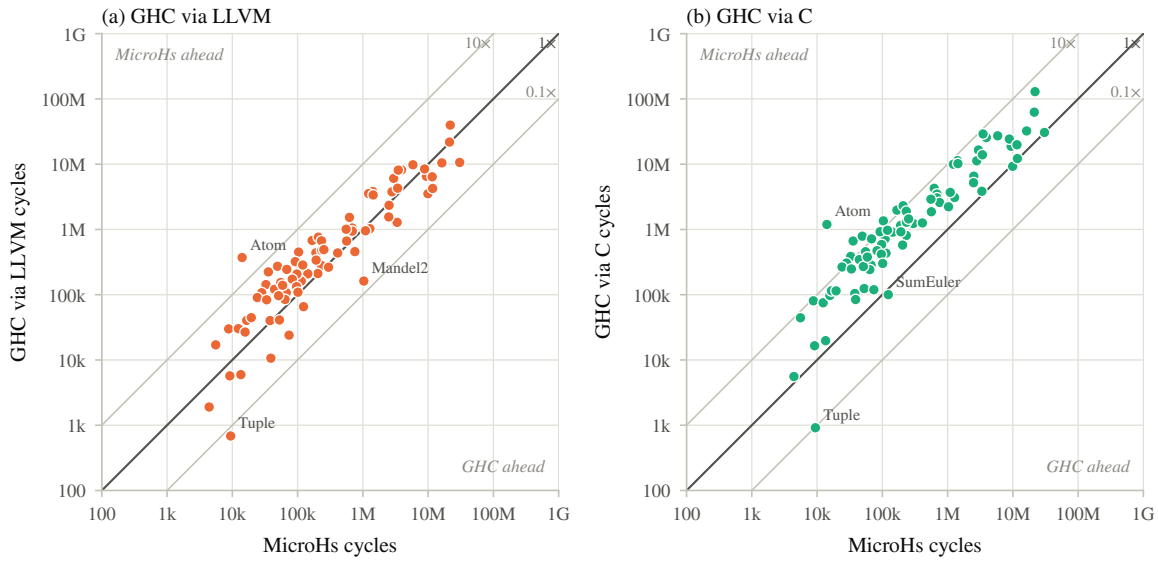


Fig. 2. GHC cycles against MicroHs cycles.

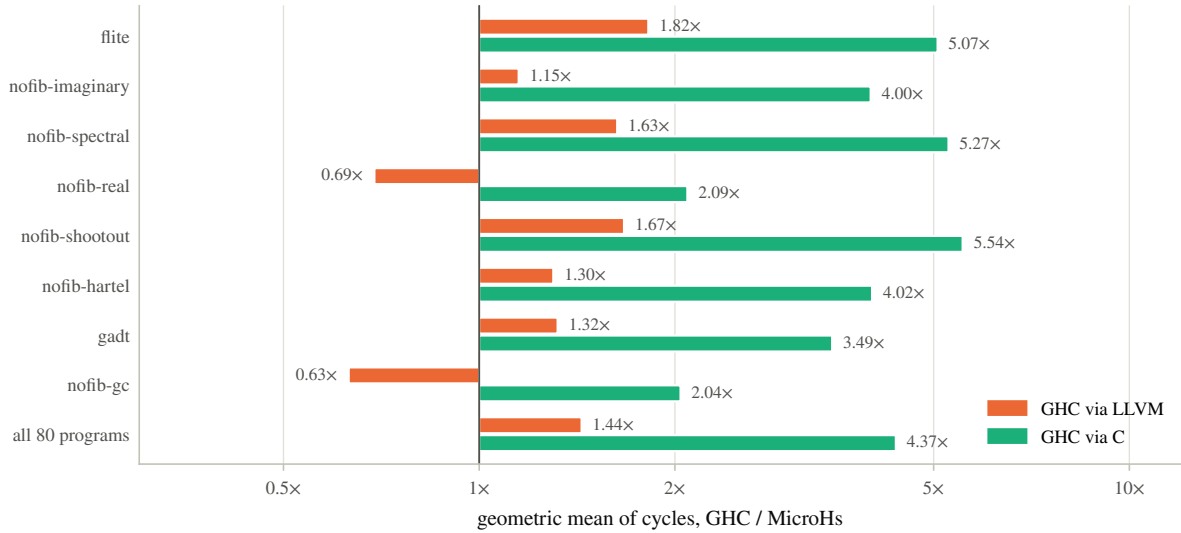


Fig. 3. Geometric mean of the cycle ratio, GHC over MicroHs, per suite.

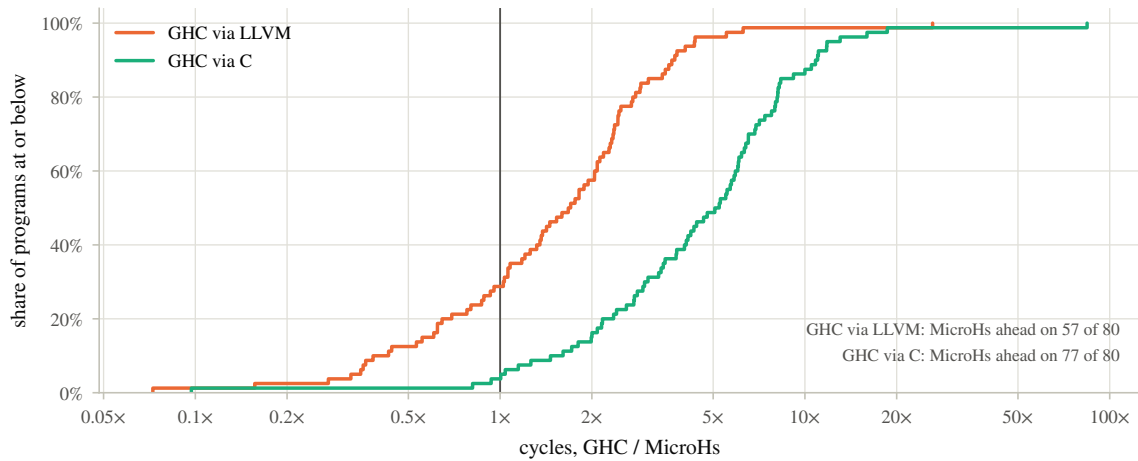


Fig. 4. Cumulative distribution of the cycle ratio, GHC over MicroHs.

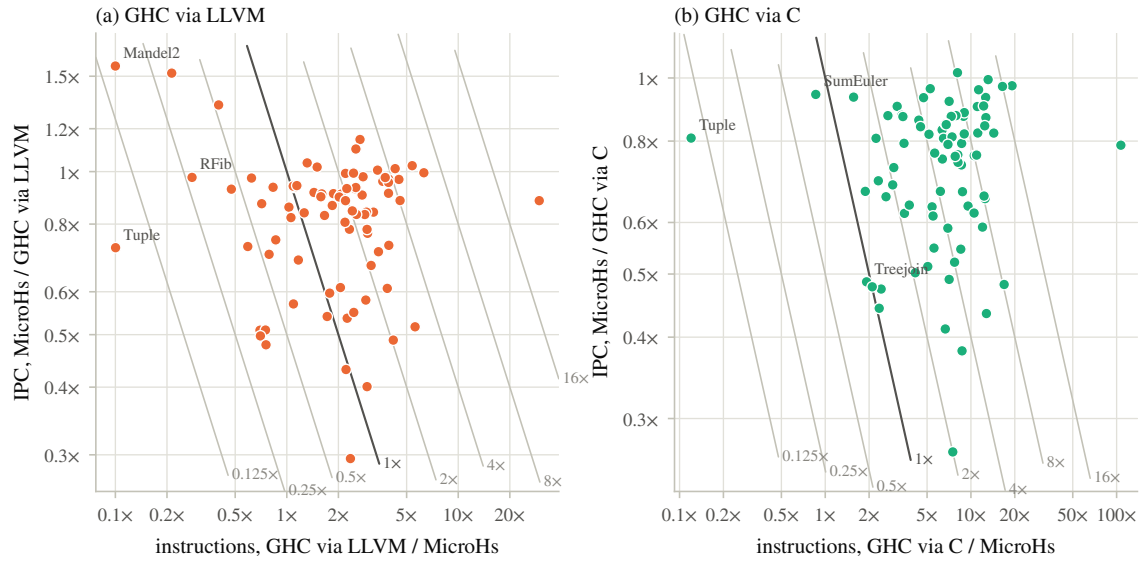


Fig. 5. IPC ratio against instruction ratio. Diagonals: equal cycle ratio.

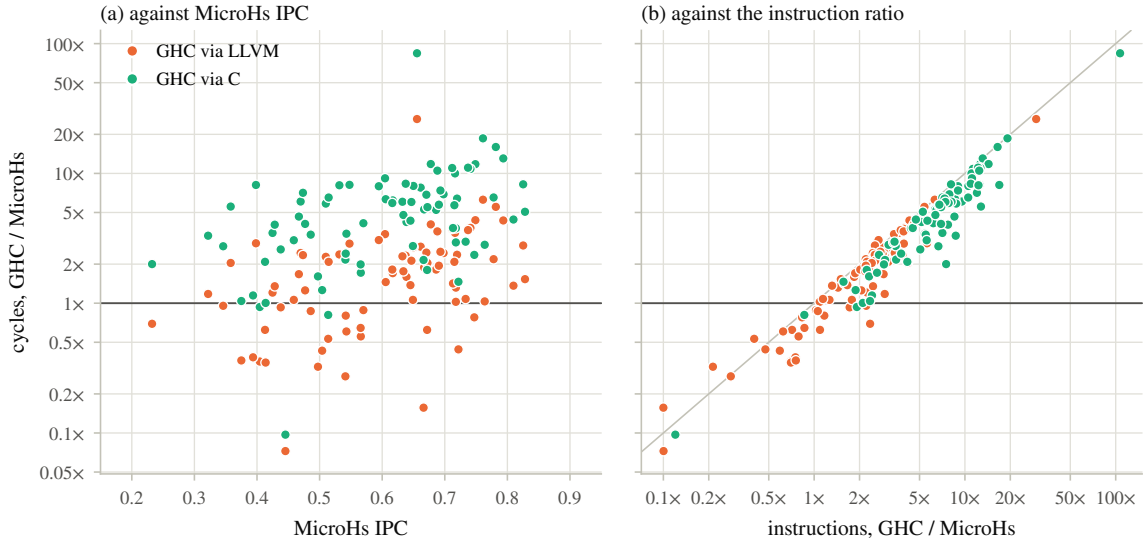


Fig. 6. Cycle ratio, GHC over MicroHs, against MicroHs IPC (a) and the instruction ratio (b).

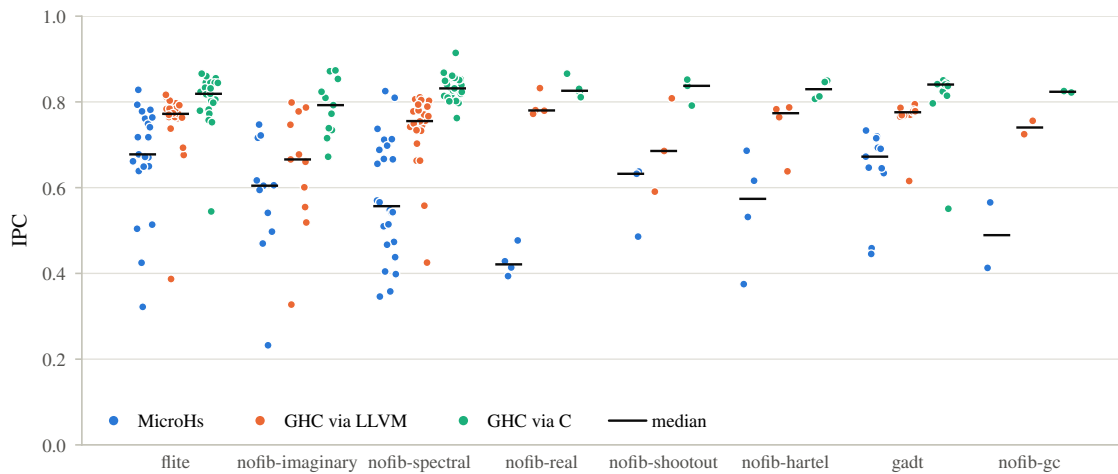


Fig. 7. IPC per suite, with medians.

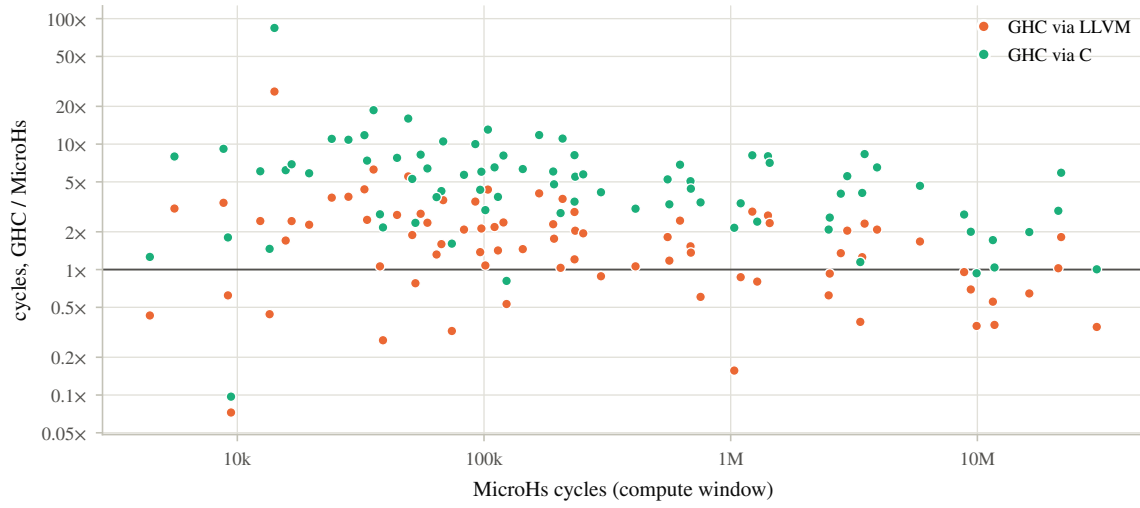


Fig. 8. Cycle ratio, GHC over MicroHs, against MicroHs cycles.

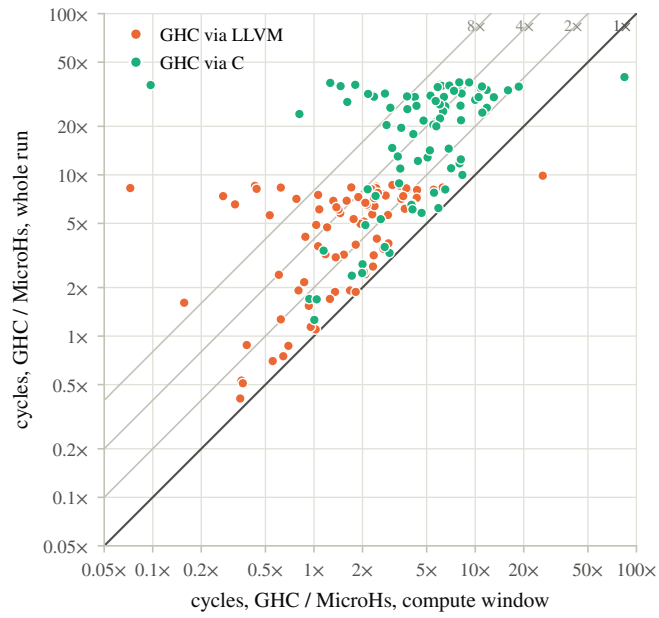


Fig. 9. Cycle ratio, GHC over MicroHs, whole run against compute window.

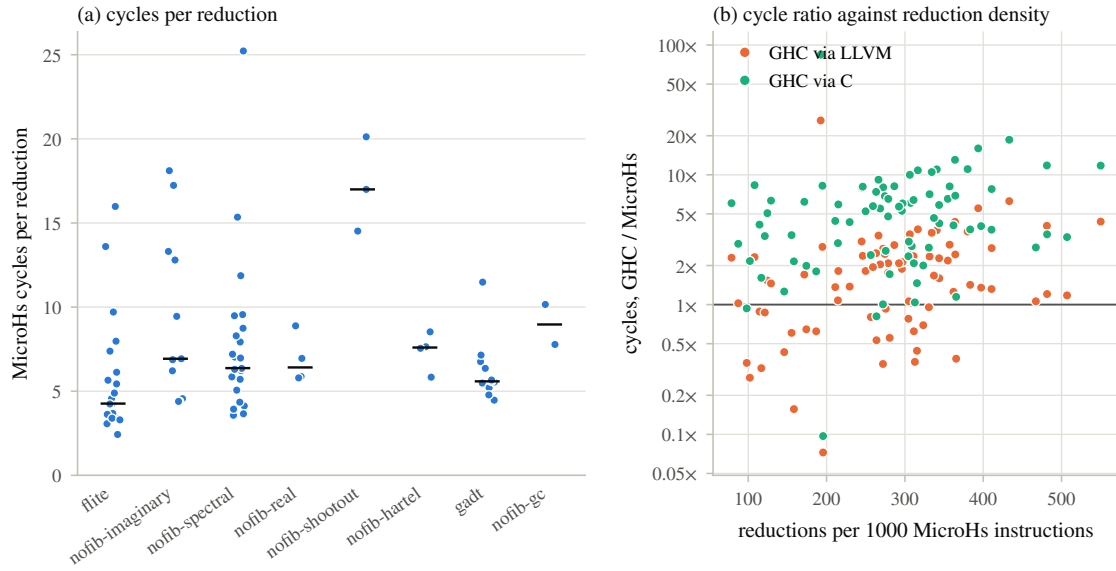


Fig. 10. MicroHs cycles per reduction per suite (a), and the cycle ratio against reductions per instruction (b).

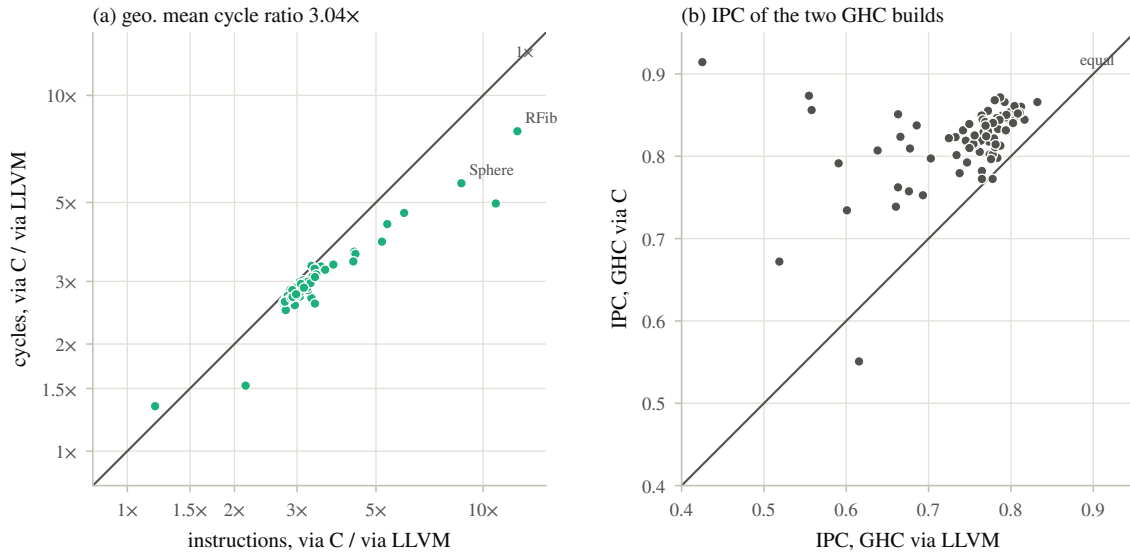


Fig. 11. GHC via C against GHC via LLVM: cycles against instructions (a), and IPC (b).

II. Caches (appendix Table II)

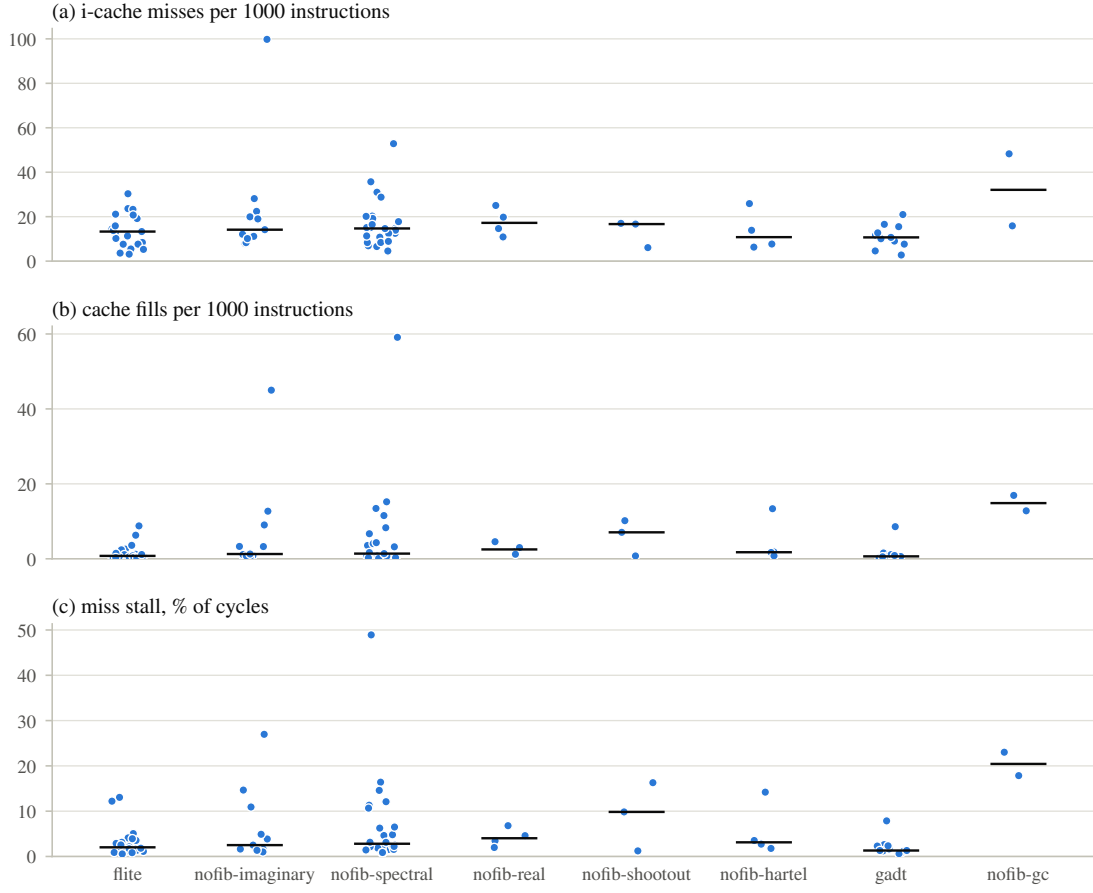


Fig. 12. MicroHs i-cache misses (a), cache fills (b) and miss stall (c) per suite, with medians.

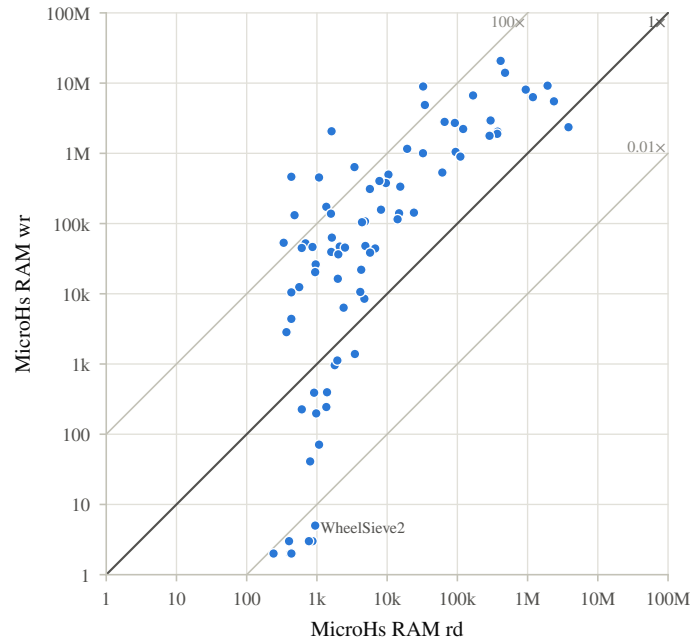


Fig. 13. MicroHs RAM writes against RAM reads.

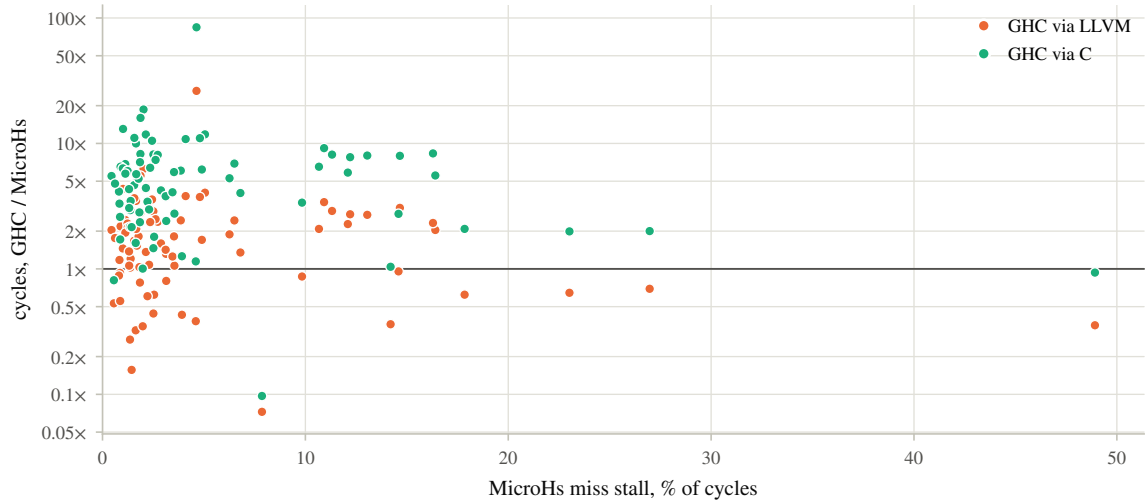


Fig. 14. Cycle ratio, GHC over MicroHs, against the MicroHs miss stall.

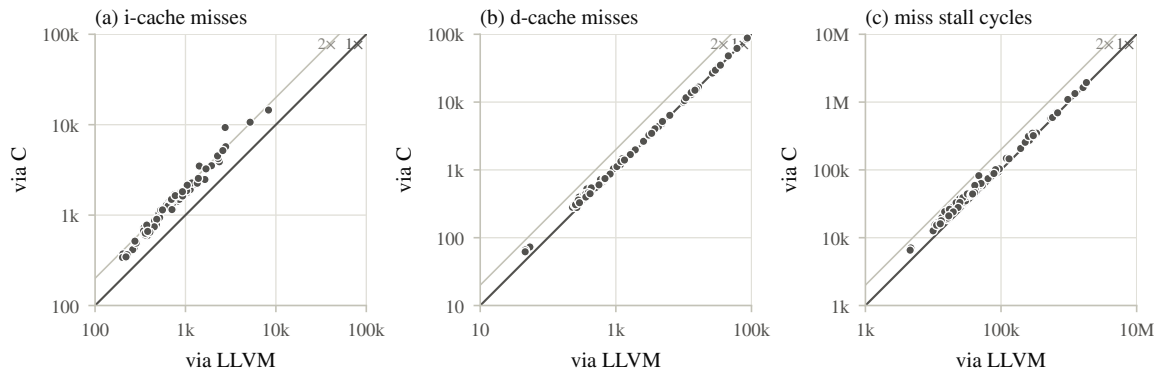


Fig. 15. GHC cache counters, via C against via LLVM.

III. Memory (appendix Table III)

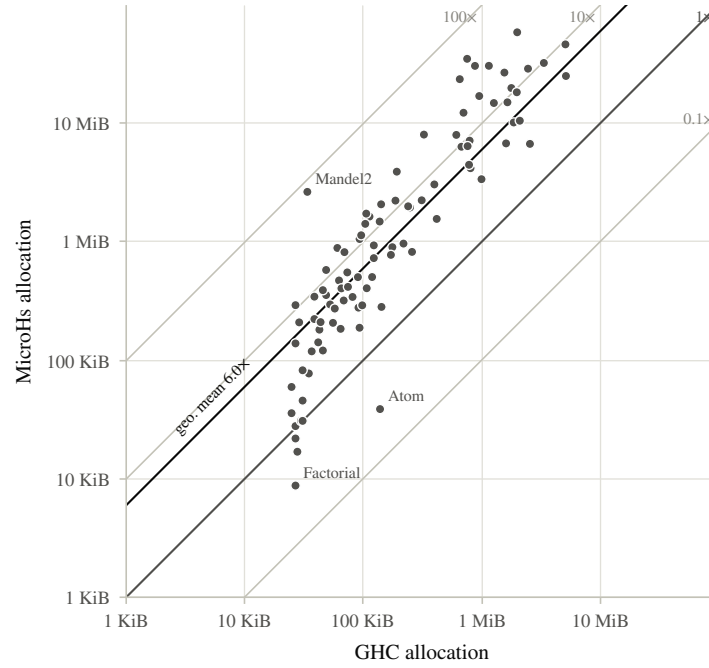


Fig. 16. MicroHs allocation against GHC allocation.

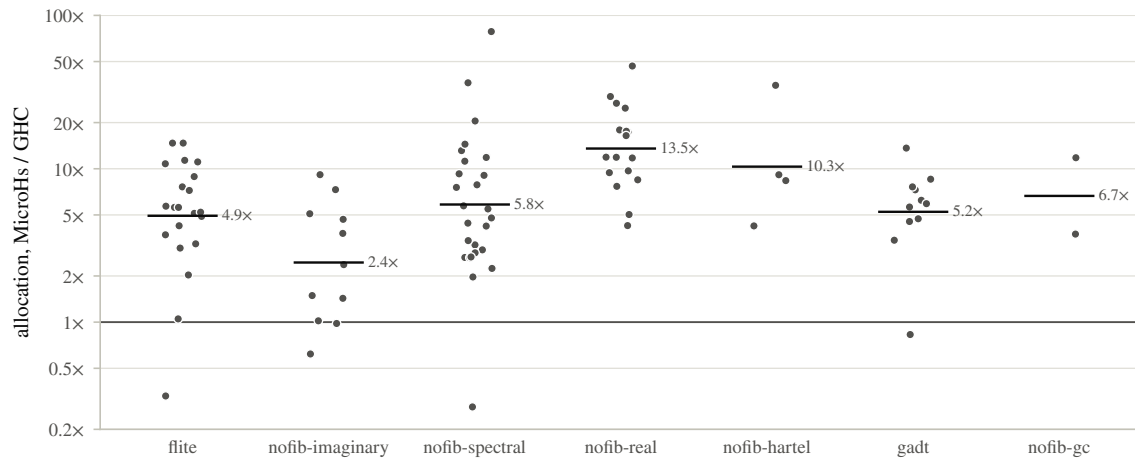


Fig. 17. Allocation, MicroHs over GHC, per suite, with geometric means.

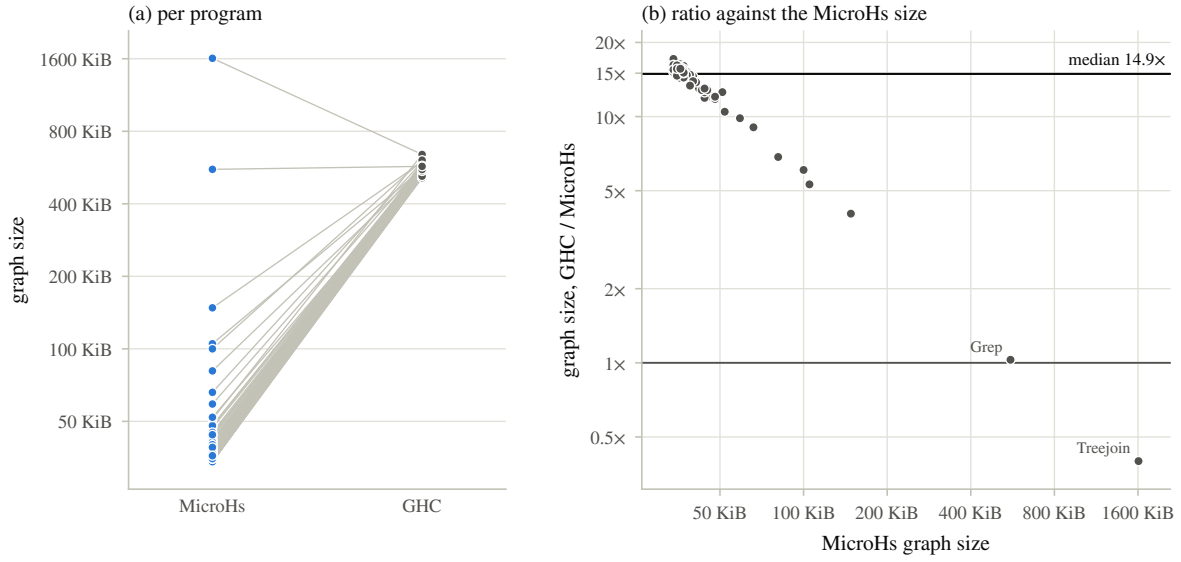


Fig. 18. Graph size of MicroHs and GHC (a), and their ratio against the MicroHs size (b).

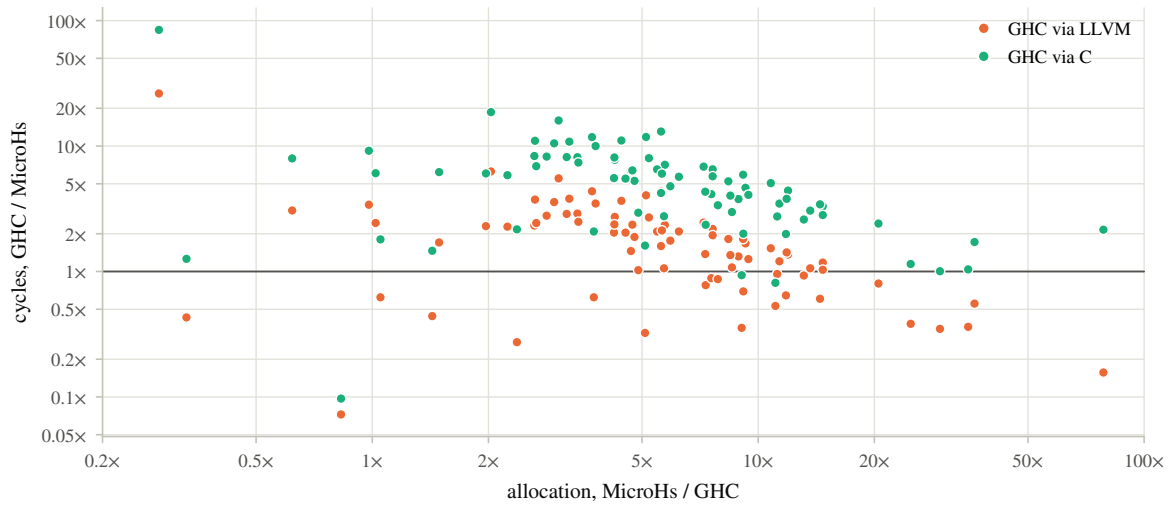


Fig. 19. Cycle ratio, GHC over MicroHs, against the allocation ratio.

IV. GHC on Blackbird-3 and on the SpaceMit M1 (appendix Table IV)

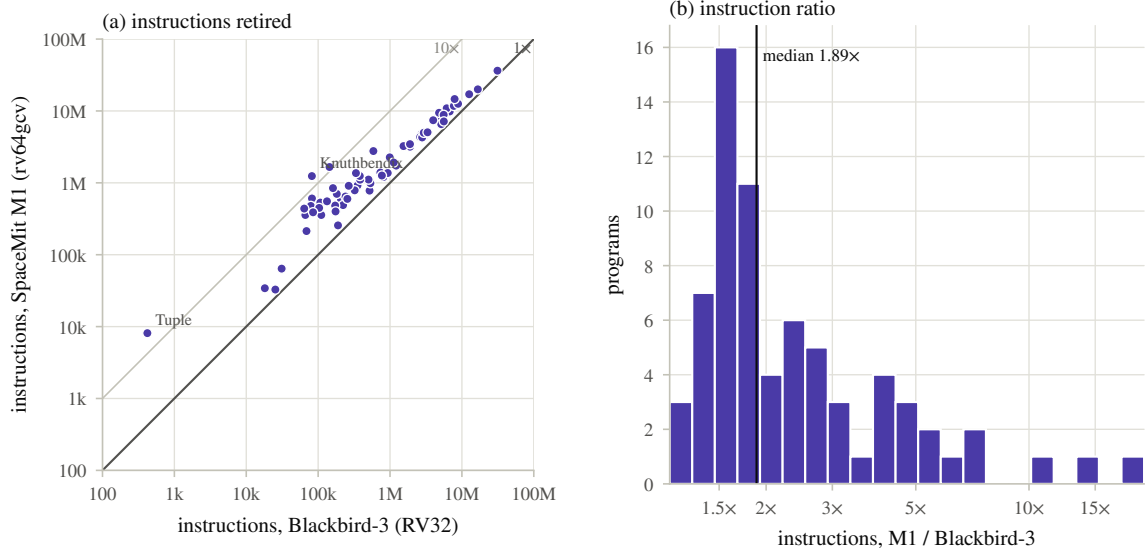


Fig. 20. Instructions retired, M1 against Blackbird-3 (a), and their ratio (b).

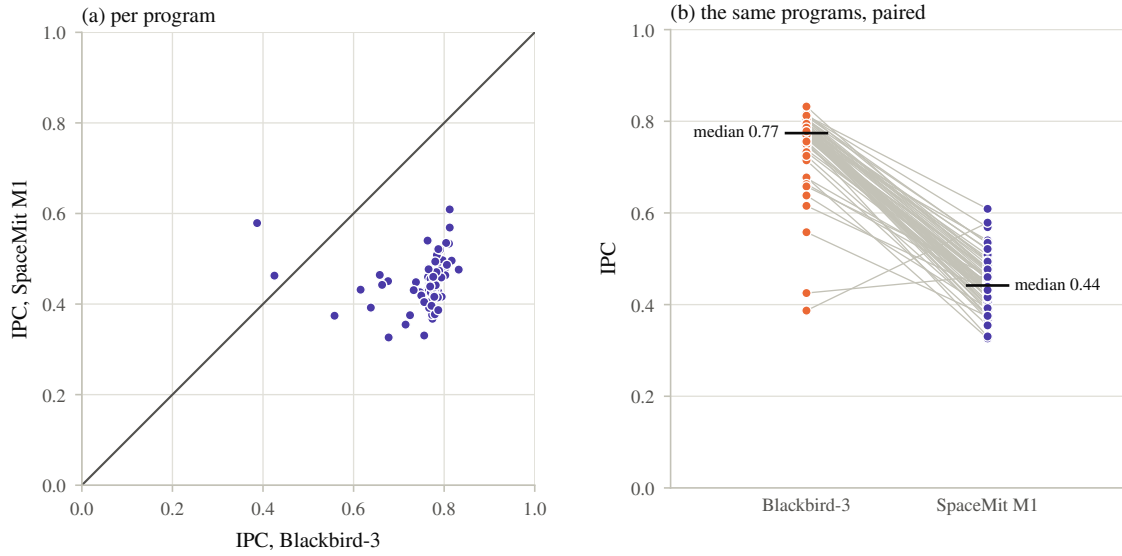


Fig. 21. IPC, M1 against Blackbird-3 (a), and paired per program (b).

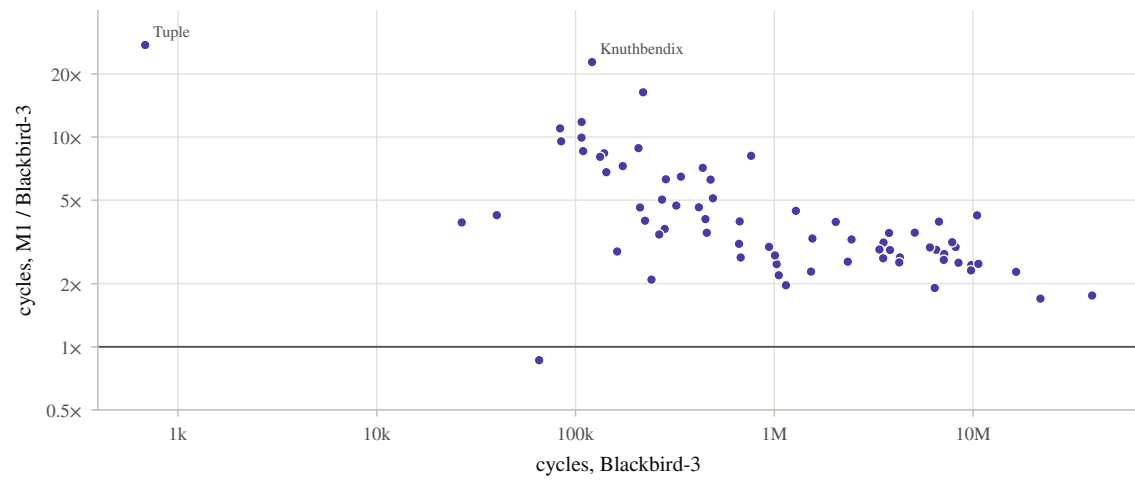


Fig. 22. Cycles, M1 over Blackbird-3, against Blackbird-3 cycles.

V. GCC, GHC and MicroHs (appendix Table V)

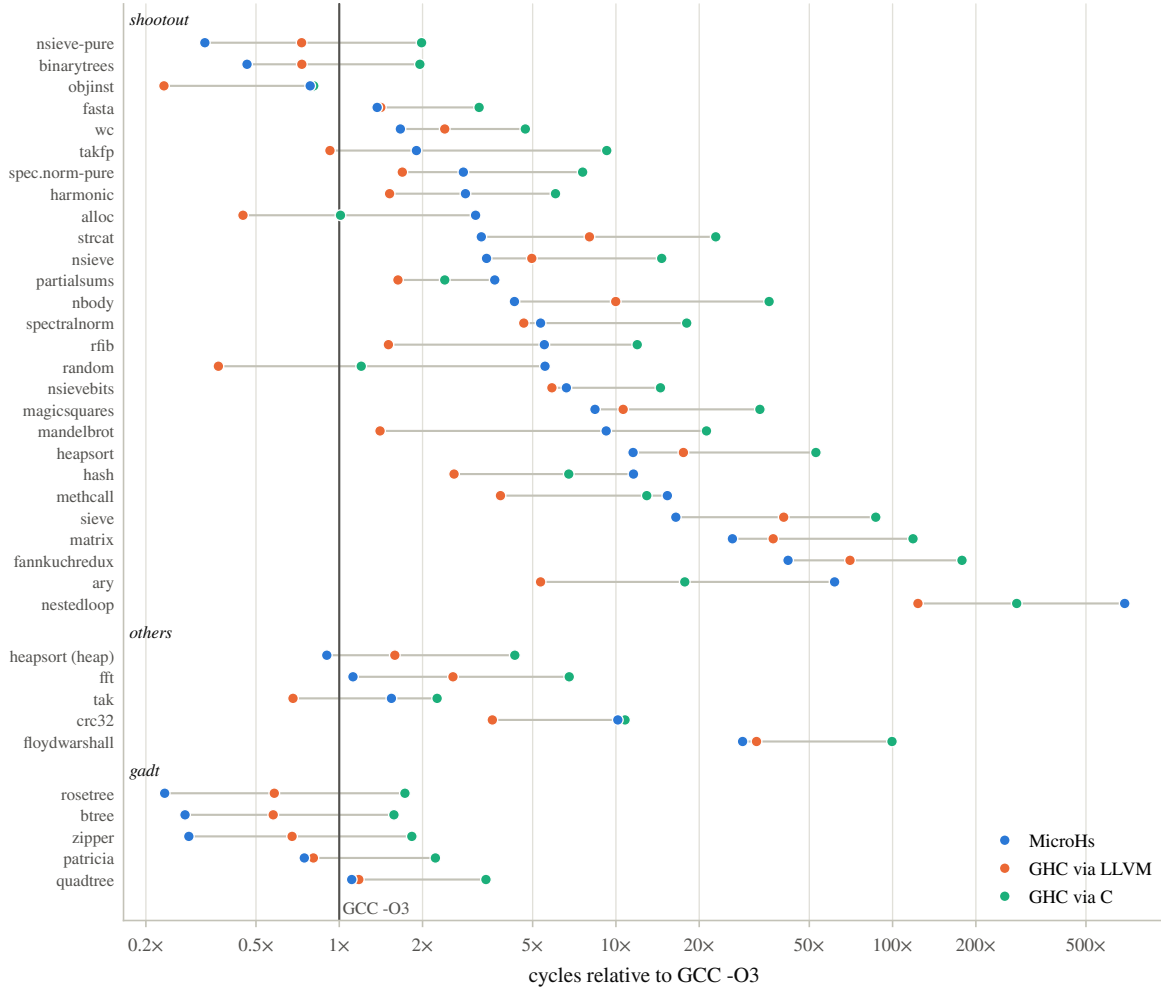


Fig. 23. Cycles relative to GCC -O3.

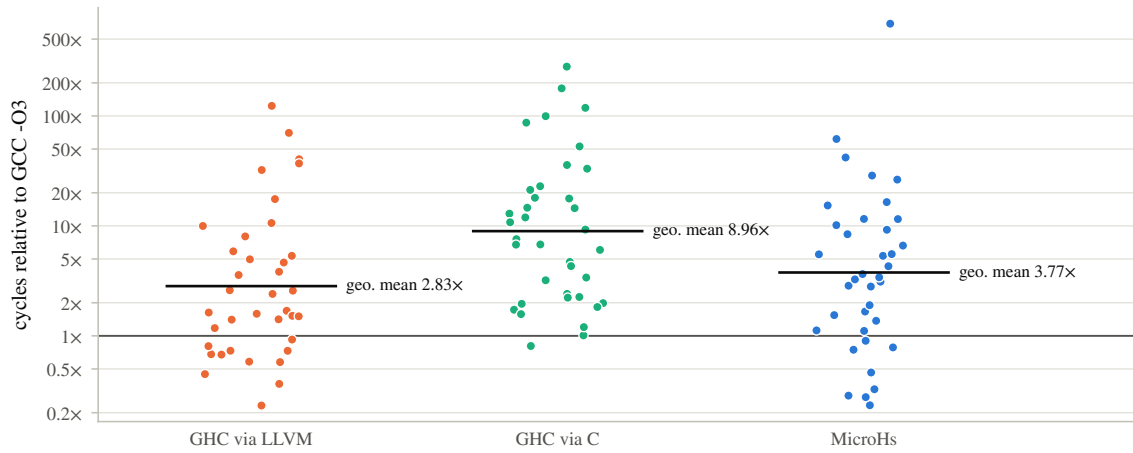


Fig. 24. Cycles relative to GCC -O3, with geometric means.

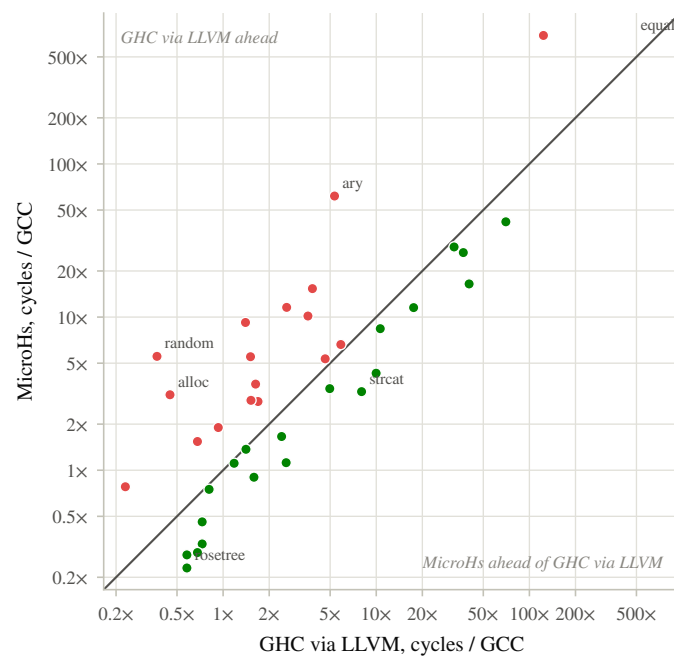


Fig. 25. MicroHs against GHC via LLVM, both relative to GCC -O3.

VI. T-cache size scan (appendix Table VI)

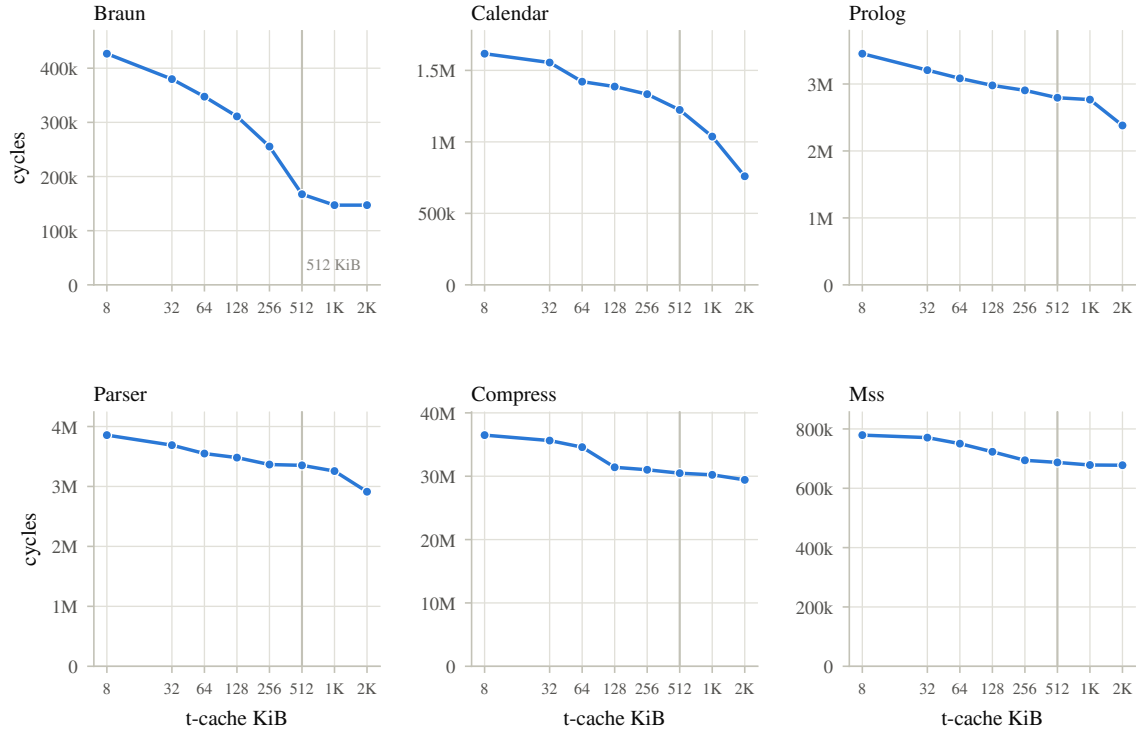


Fig. 26. Cycles against t-cache size.

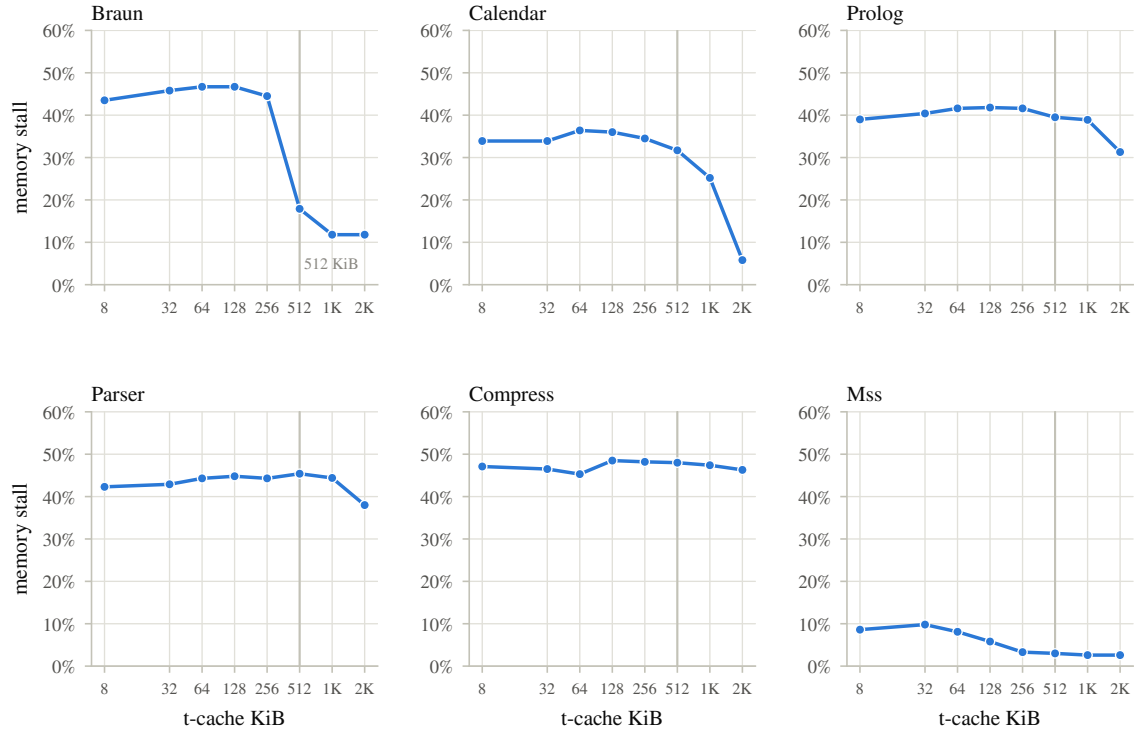


Fig. 27. Memory stall against t-cache size.

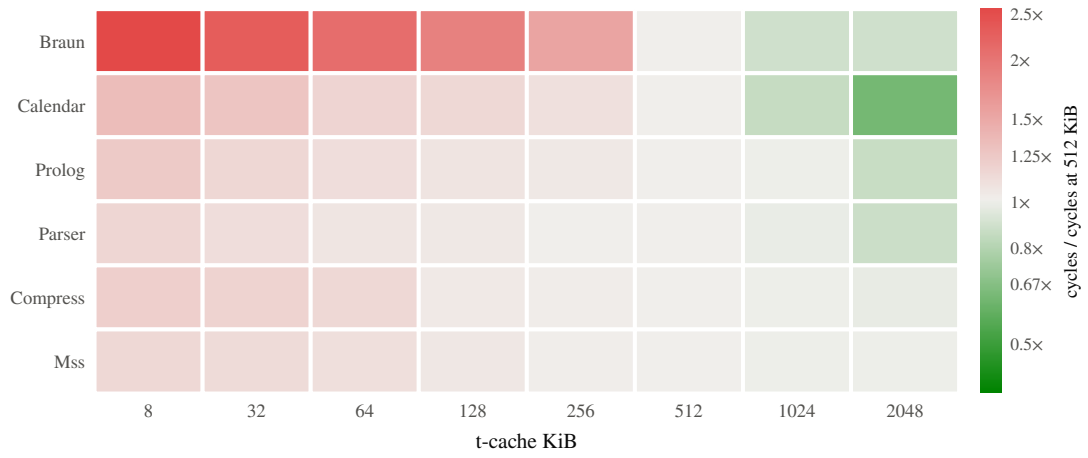


Fig. 28. Cycles relative to 512 KiB.

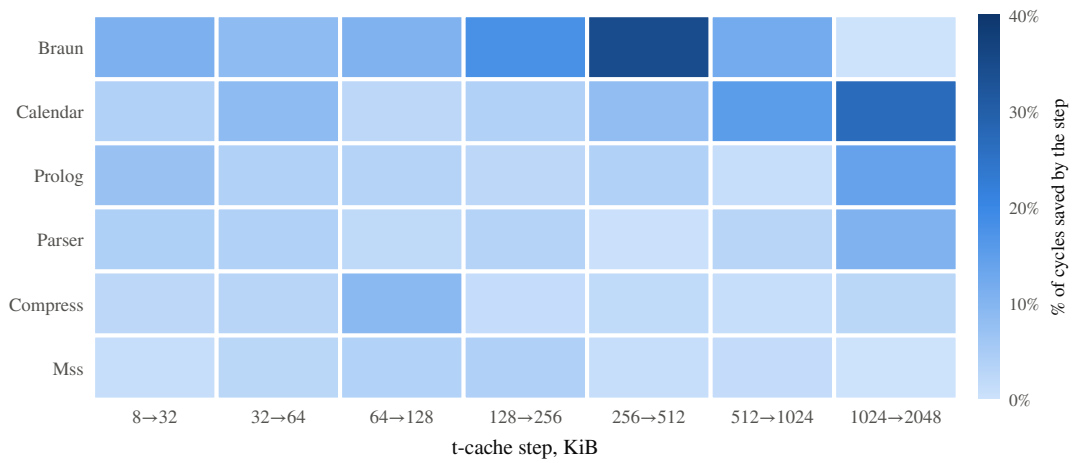


Fig. 29. Cycles saved by each t-cache step.

VII. PMC breakdown (appendix Table VII)

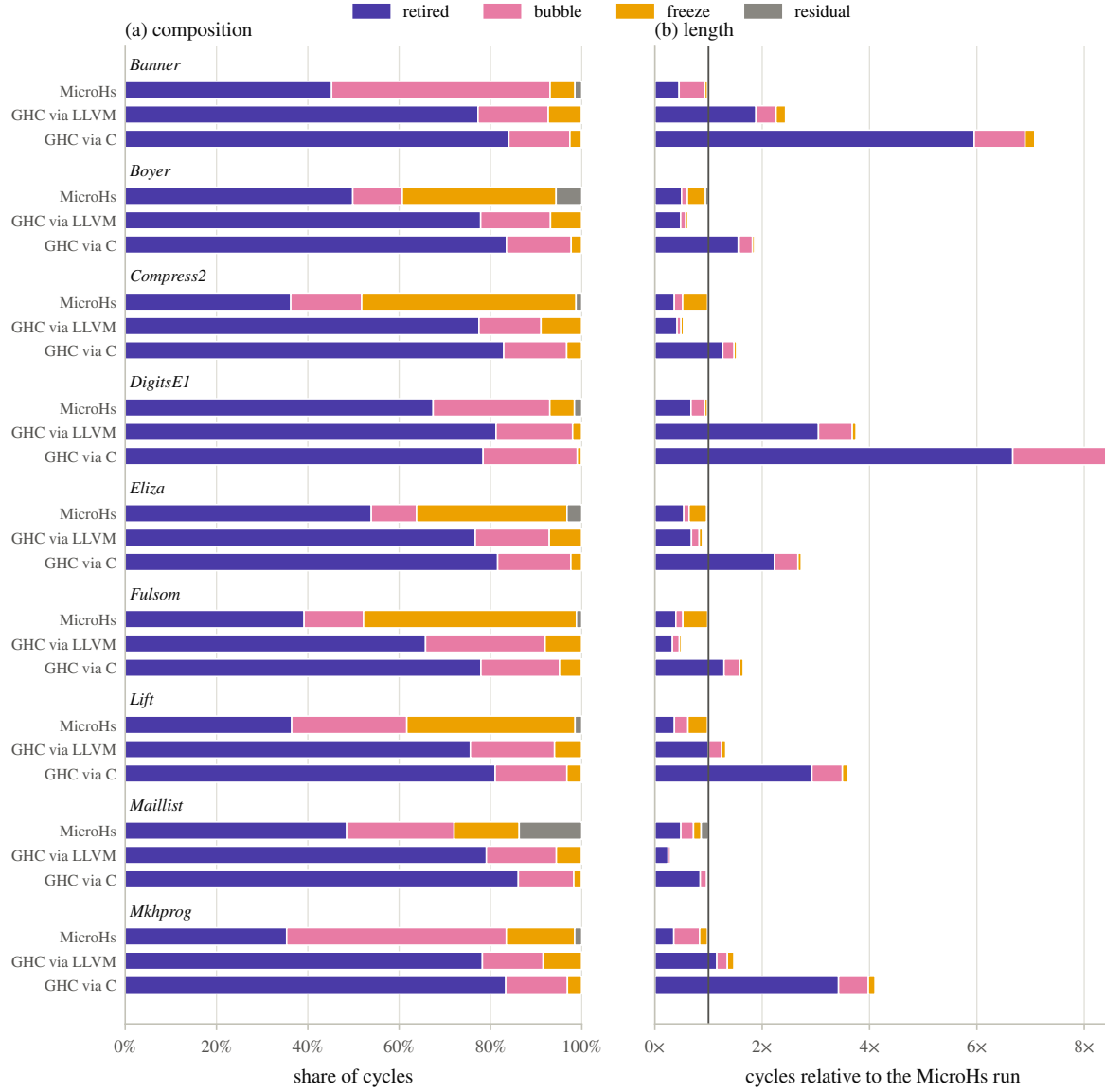


Fig. 30. Cycle split per program and build, as shares (a) and relative to MicroHs (b).

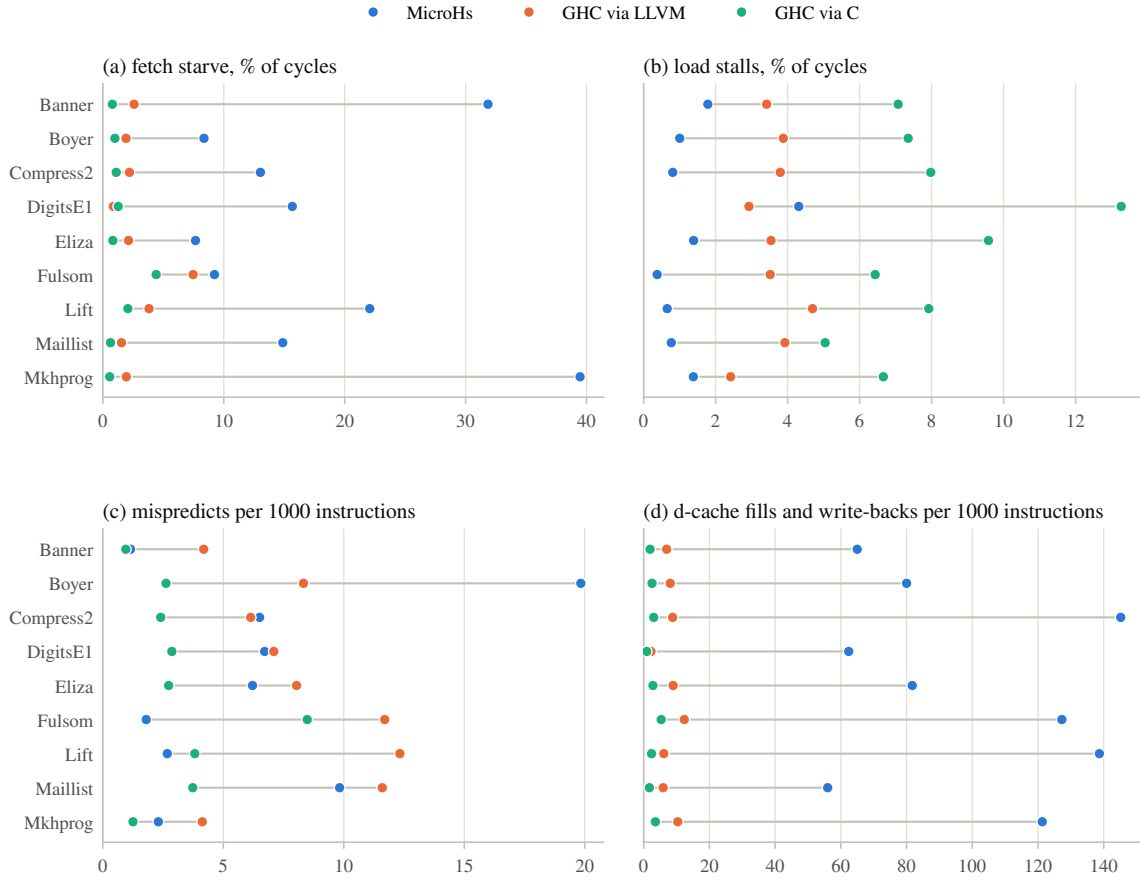


Fig. 31. Stall and event rates per program and build.

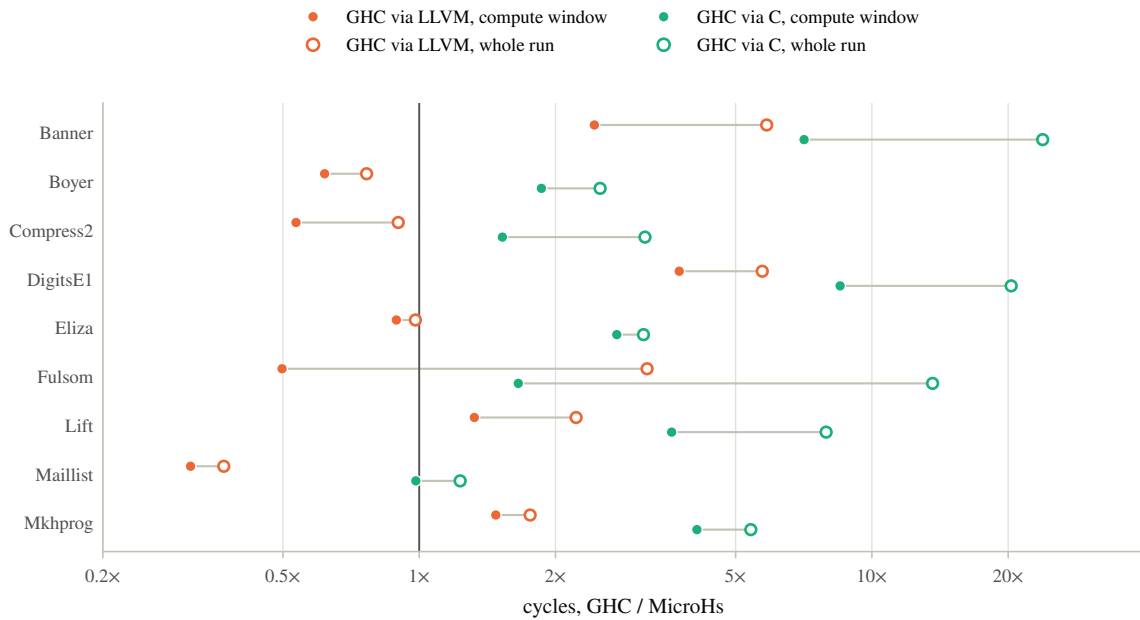


Fig. 32. Cycle ratio, GHC over MicroHs, compute window and whole run.

VIII. Cycle breakdown (appendix Table VIII)

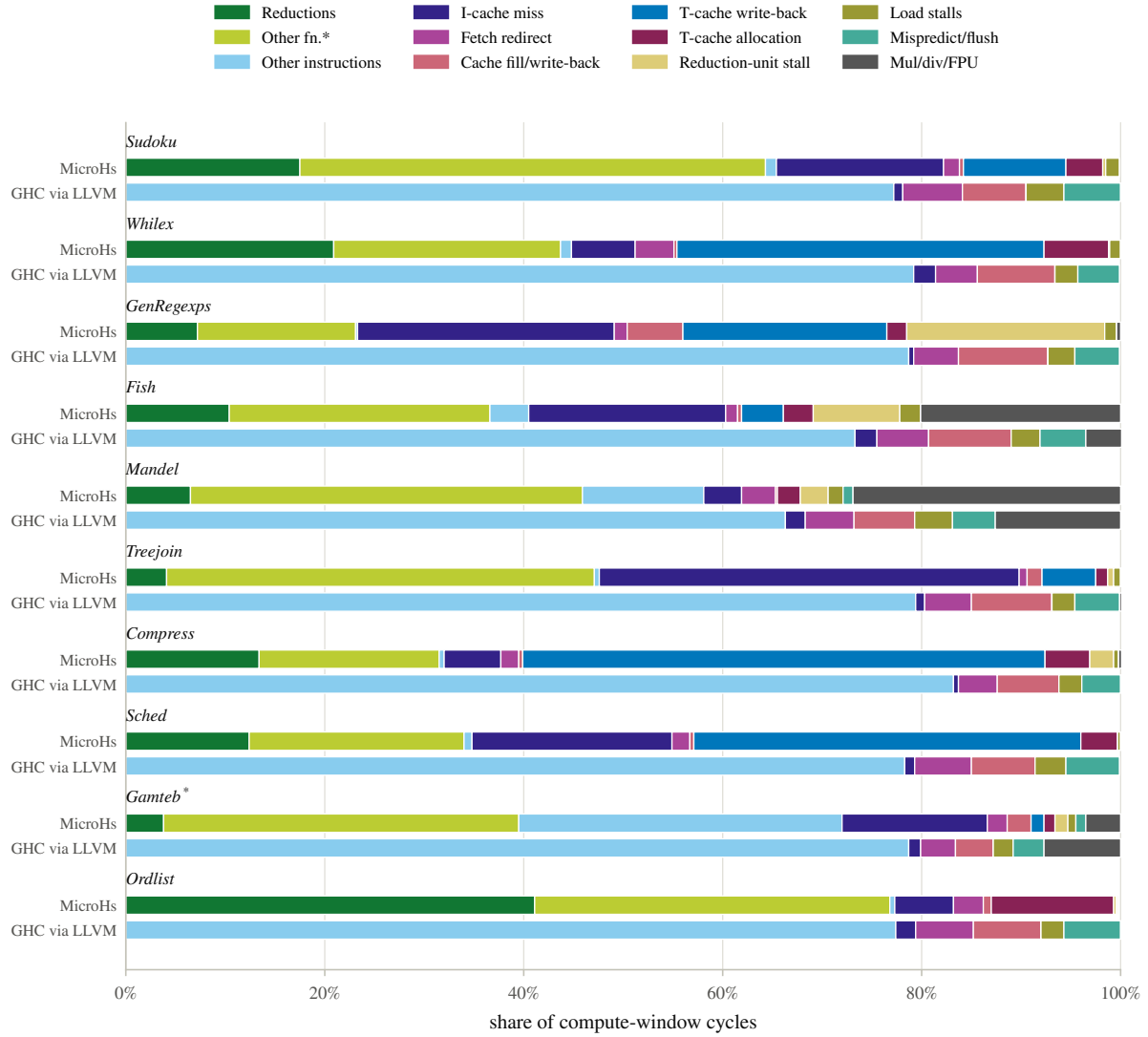


Fig. 33. Cycle breakdown per program.

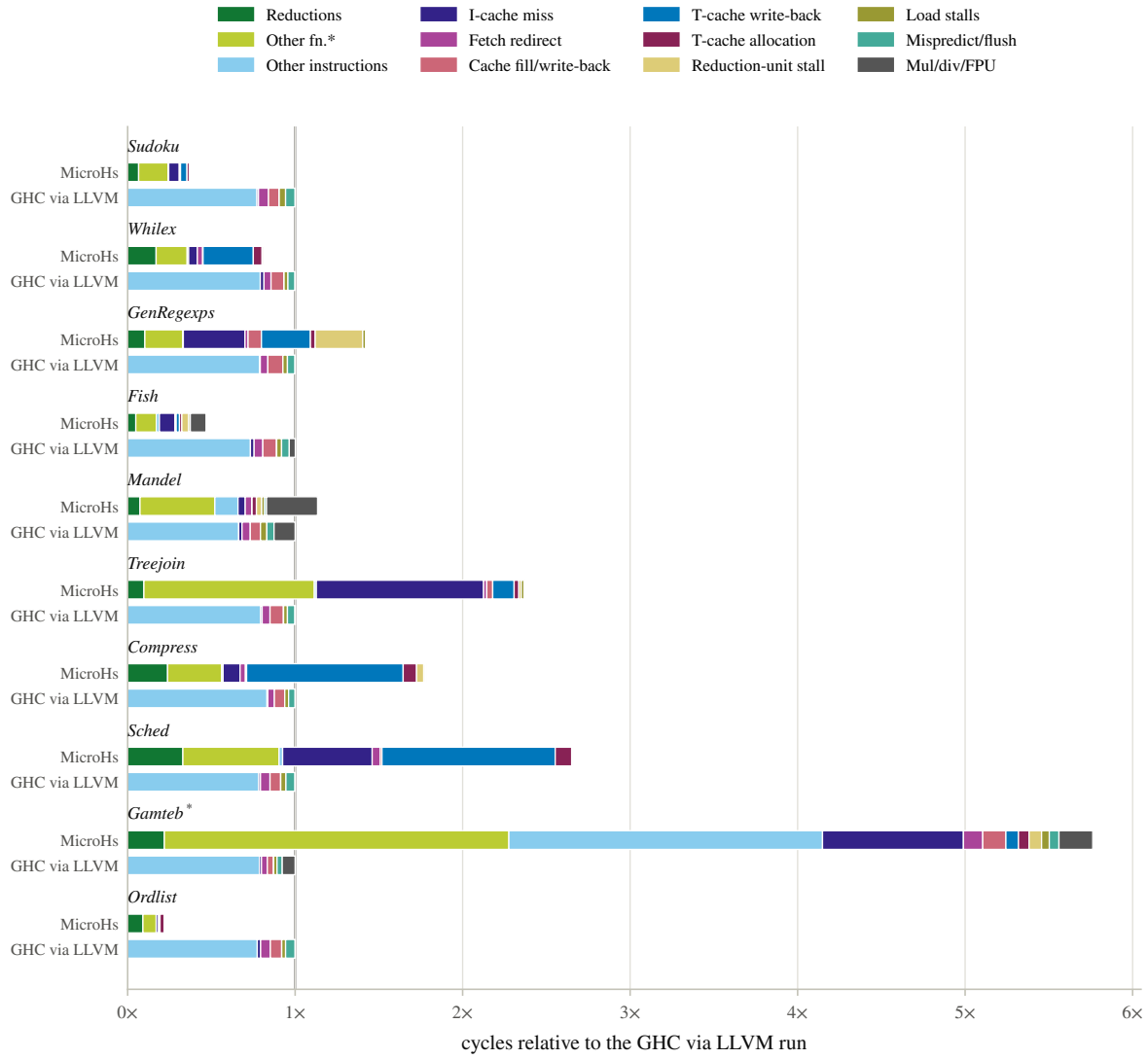


Fig. 34. Cycle breakdown per program, relative to GHC via LLVM.

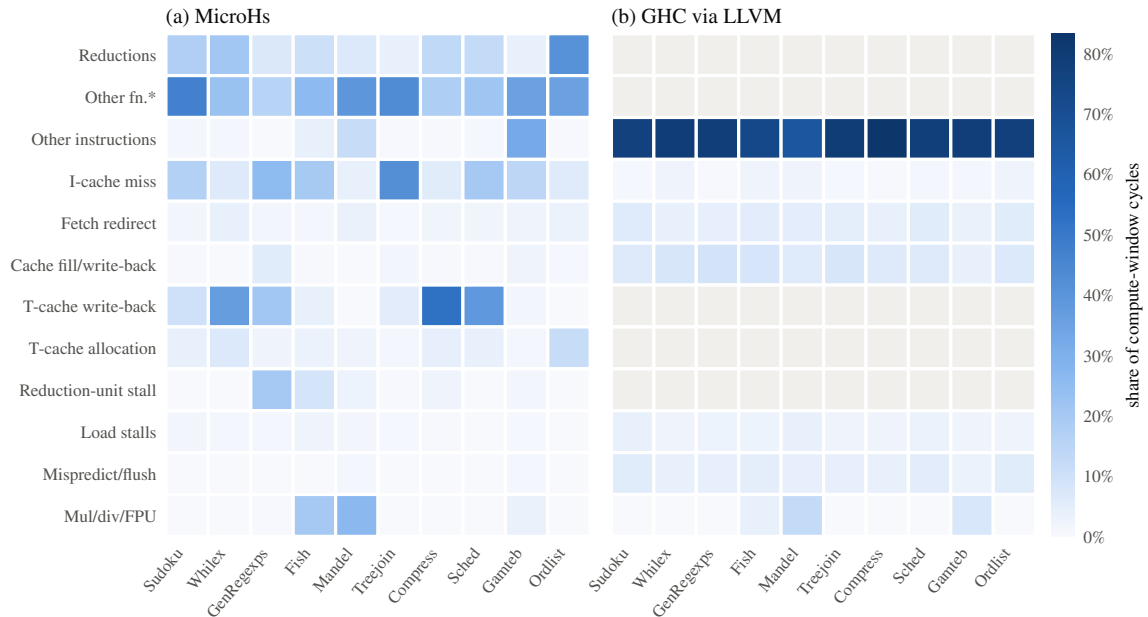


Fig. 35. Share of cycles per class and program.

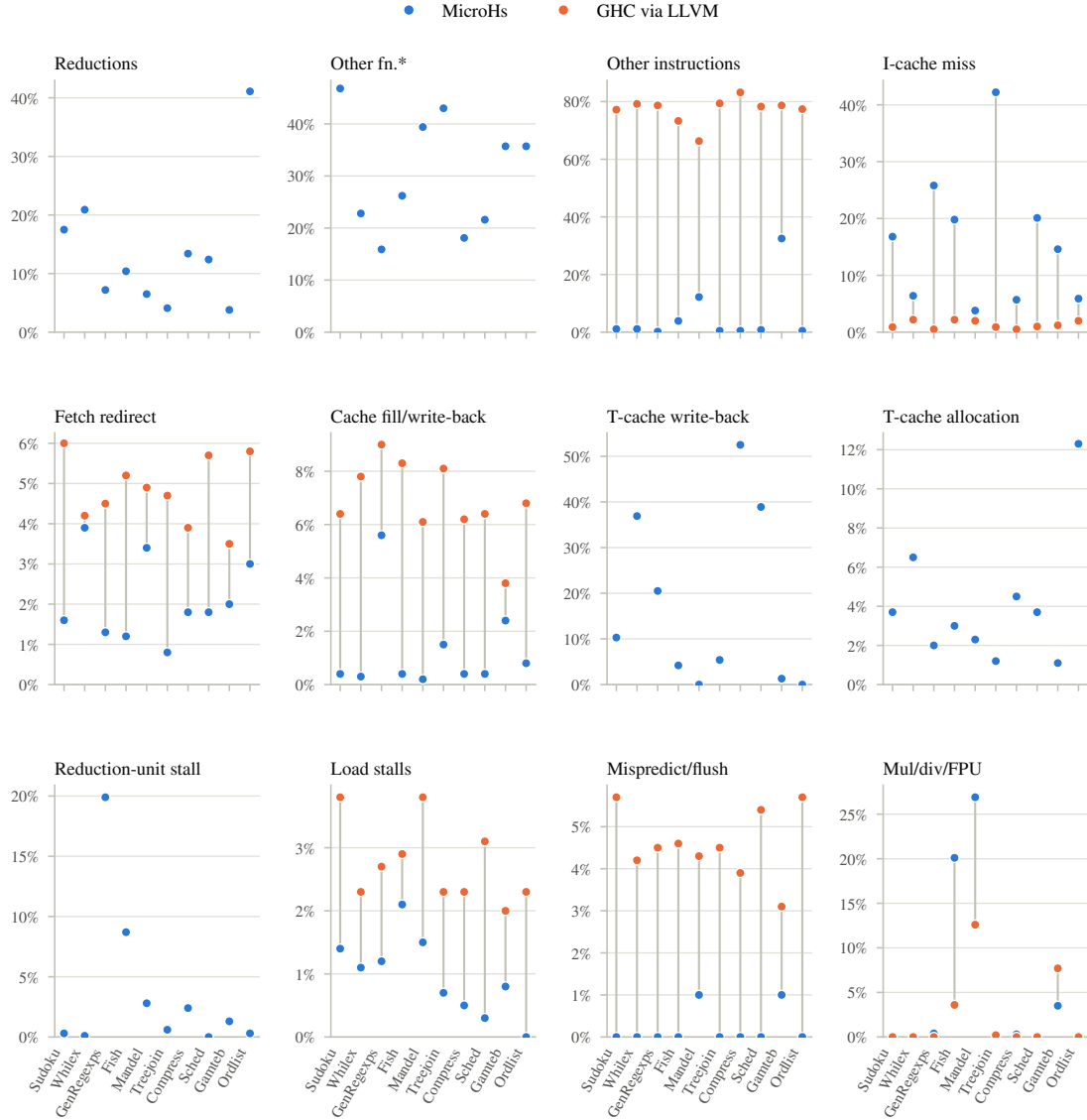


Fig. 36. Share of cycles per program, one panel per class.

IX. Embench-IoT (appendix Table IX)

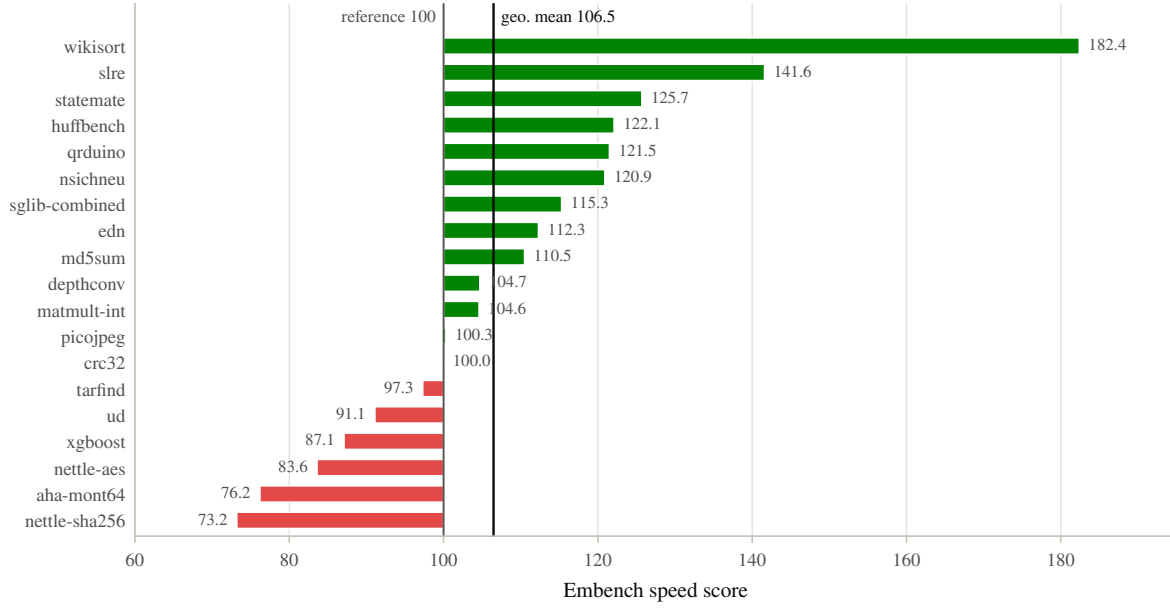


Fig. 37. Embench speed score per benchmark.

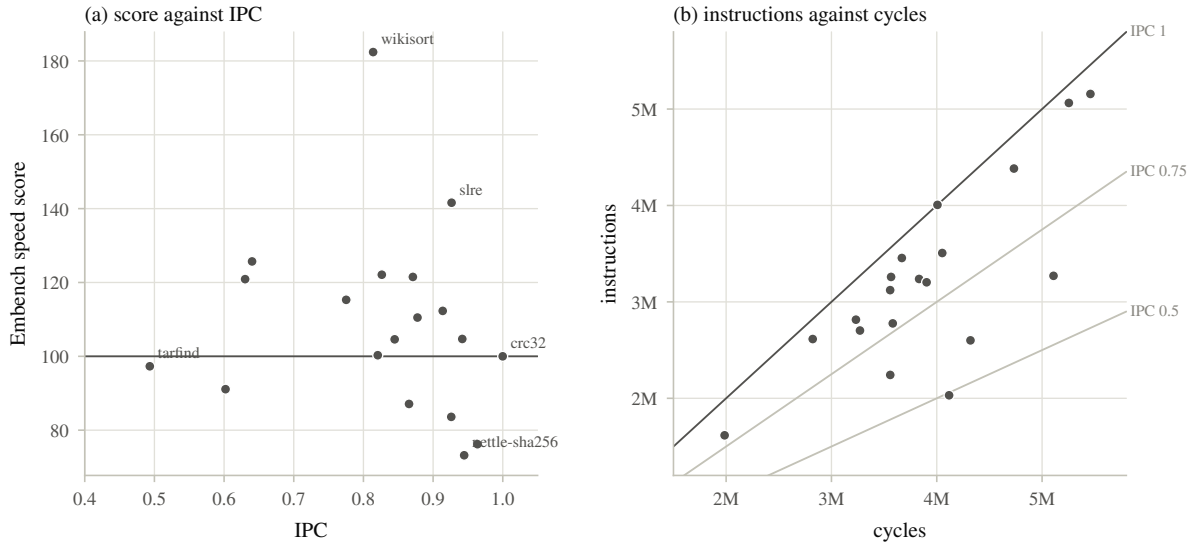


Fig. 38. Score against IPC (a), and instructions against cycles (b).